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Improved statistical multi-scale analysis of fractured reservoir analogues

Vincenzo Guerriero *, Stefano Vitale, Sabatino Ciarcia, Stefano Mazzoli

Dipartimento di Scienze della Terra, Università di Napoli 'Federico II,' Largo San Marcellino 10, 80138 Napoli, Italy

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ABSTRACT

This paper includes the results of the application of novel methodologies for the statistical analysis of tensile fracture attributes (spacing, density, and aperture). The statistical methods have been applied to a fractured Lower Cretaceous carbonate succession cropping out in the Sorrento Peninsula (Naples area, Italy), representing a geological analogue of buried reservoirs of southern Apennines major oil fields. As fracture networks (observable from outcrop to the crystal scale) can significantly affect the hydraulic behavior of a fractured reservoir, the definition of appropriate models of fracture spacing and aperture probability distribution at various scales represents an important goal of structural analysis. In order to achieve such an objective, sampling has been carried out at different scales, by means of traditional scan lines at outcrop, as well as micro-scan lines carried out both by a digital micro-camera (50×) and by the optical microscope (200×). Fracture spacing and density analysis are mainly aimed at establishing the characteristics of joint spatial distribution over a range of scales. Aperture analysis is aimed at verifying that the aperture cumulative frequency is described by a power law and, in the latter instance, to provide an effective method for a more precise determination of the exponent of the power law. Finally, new expressions are provided, in closed form, in order to calculate the confidence interval of the power-law exponent. The proposed criteria provide handy and effective methods for a significantly improved statistical analysis of fracture attributes.

1. Introduction

Most of the models describing the hydraulic behavior of fractured reservoirs assign the main contribution of rock permeability to fracture systems (e.g. Warren and Root, 1963). In fractured bedded rocks, such as shallow-water limestones, several fracture sets may occur, each one being characterized by a different statistical behavior. Based on a wide data set from reservoir analogues, Odling et al. (1999) pointed out the occurrence of two main types of tensile fracture systems (Fig. 1): (i) stratabound systems, characterized by fractures terminating along the margins of mechanical layers, and (ii) non-stratabound systems, formed by fractures displaying lengths lesser than bed thickness, or fractures crosscutting more than one mechanical layer. Stratabound fractures generally show regular spacing, which is well documented as being proportional to bed thickness (Bai and Pollard, 2000; Gross, 1993; Gross and Engelder, 1995; Huang and Angelier, 1989; Mandal et al., 1994; Narr, 1996; Narr and Suppe, 1991; Odling et al., 1999; Pascal et al., 1997; Price, 1966; Wu and Pollard, 1995). On the contrary, non-stratabound systems show irregular spacing, and commonly exhibit a statistical distribution of fracture aperture values that is best described by a power law over several scales of observation (Belfield, 1994; Belfield and Sovich, 1995; Guerriero et al., 2010; Ortega et al., 2006). It should be emphasized that a power-law distribution never attains a mean value that is finite and different from zero (this can be easily verified by integration).

Taking into account Odling et al.'s (1999) classification of joints, several permeable systems, characterized by various features and different scales, may be distinguished in building up a rock permeability model. At the meter/decimeter scale stratabound fractures form, together with bedding-parallel layer joints, a permeable network able to convey fluids toward more permeable higher-scale systems (fault damage zones). At the centimeter- to crystal-scale, non-stratabound fractures form a pervasive and capillary network whose permeability mainly depends on features such as connectivity. The latter, in turn, depends on fracture density and length distribution; furthermore, it varies with the scale of observation (Davy et al., 2006; Odling, 1997; Odling et al., 1999). Taking into account that these fractures occur at several scales (Belfield, 1994; Belfield and Sovich, 1995; Bonnet et al., 2001; Guerriero et al., 2010; Ortega et al., 2006) down to the crystal scale (true), fracture density can attain extremely high values (even of thousands of fractures per meter). Consequently, connectivity may become very high. Higher values of fracture density refer to smaller fractures that may often be filled due to mineralization (Laubach, 2003). Nevertheless, even in case tiny joints are occluded by cement — and therefore cannot contribute directly to reservoir permeability — these fractures are useful for predicting the frequency of larger fractures. From this viewpoint, fracture aperture distribution

* Corresponding author. Tel.: +39 0812538124.
E-mail address: vincenzo.guerriero@unina.it (V. Guerriero).

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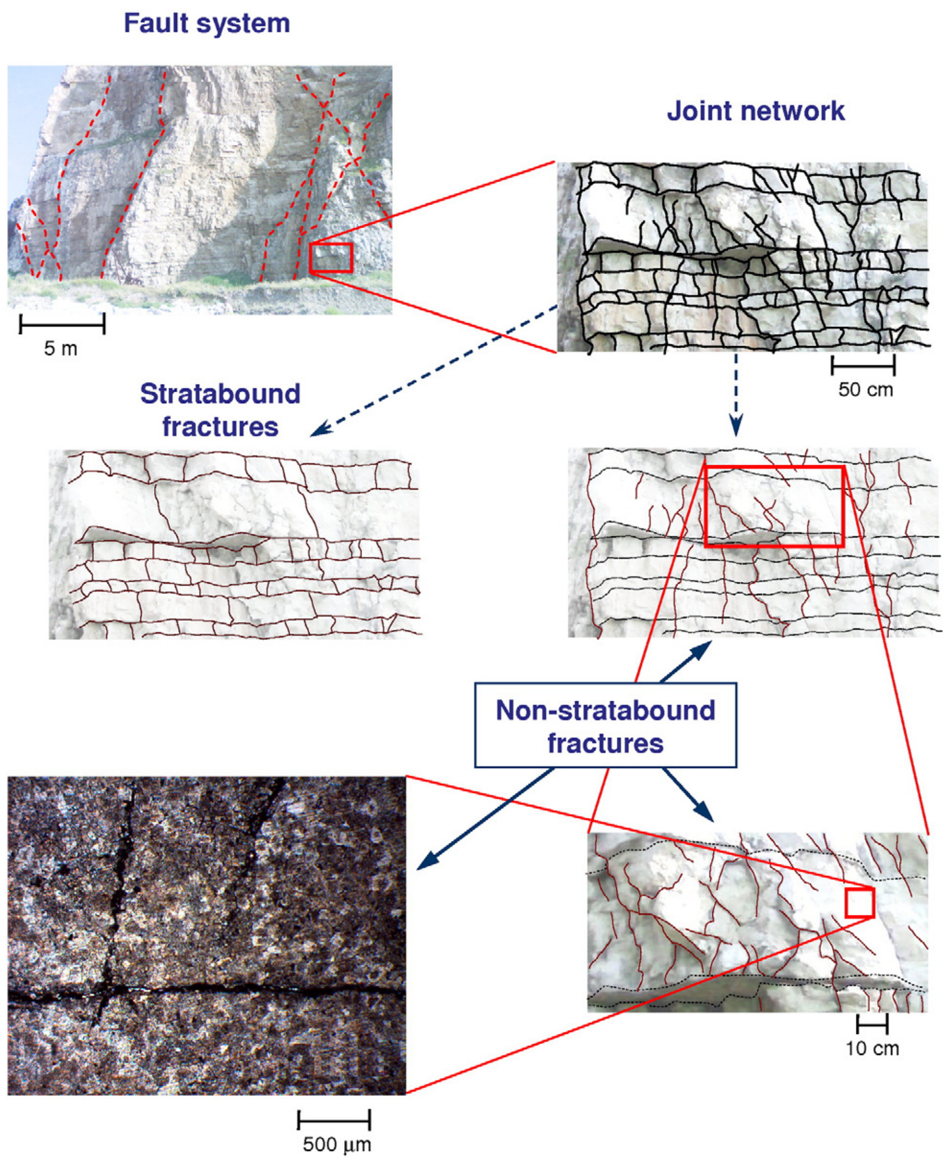

Fig. 1. Permeable systems in rocks at various scales of observation.

assumes a key role in evaluating permeability (and porosity) of fracture networks. According to this model, the non-stratabound fracture network attains a crucial role in the hydraulic response to fluid (hydrocarbons, water) extraction. Firstly because, in case it is highly pervasive, it can provide a secondary porosity contribution comparable to the primary one, and hence such a system could contain an important amount of fluid stocked in the reservoir; this fluid could be released over a notably shorter time than that present in the pore system of the non-fractured host rock (which is characterized by a much lower permeability). Secondly, because joint set 'pervasiveness' and maximum density could substantially affect the reservoir response to a non-steady-state flow. Indeed the time spent by fluid particles flowing throughout the porous matrix (whose permeability could be several orders of magnitude lower than that of the fracture network) is proportional to the average path from the pores to the fracture network, which is equal to one half of the (true) mean spacing.

Useful and handy statistical analysis methods, aimed at estimating aperture distribution and related uncertainties, are needed taking into account the noticeable difficulties in recording accurate aperture measurements and in performing adequately sized scan lines over a range of observation scales. On the other hand, only few works dealing with aperture statistical analysis are available in the literature.

The main purpose of this paper is to illustrate an effective methodology of statistical multi-scale analysis aimed at assessing the probability distribution of fracture apertures, spacing and related density. In particular, our research is aimed at: (i) characterizing the type of aperture probability distribution and comparing it with published models (e.g. Ortega et al., 2006); (ii) defining the type of probability distribution of spacing and the inverse of mean spacing (i.e. fracture density); and (iii) estimating the main statistical parameters (e.g. mean value, variance, exponent of power-law distributions) and quantifying related uncertainties. This methodology involves the use of statistical diagnostic tools such as probability plots and autocorrelation analysis, traditionally employed in other scientific contexts, as well as novel criteria of statistical analysis of fracture spacing and aperture.

The methods have been applied to a fractured Lower Cretaceous carbonate succession cropping out in the Sorrento Peninsula (Naples area, Italy, Fig. 2), already investigated using a more traditional scheme of analysis (Guerriero et al., 2010), in order to better characterize fracture network features. The studied succession represents a geological analogue of the buried reservoirs of the southern Apennines oil fields (Iannace et al., 2008; Shiner et al., 2004; Corrado et al., 2002). The studied carbonate strata belong to the Triassic–Cenozoic shallow-

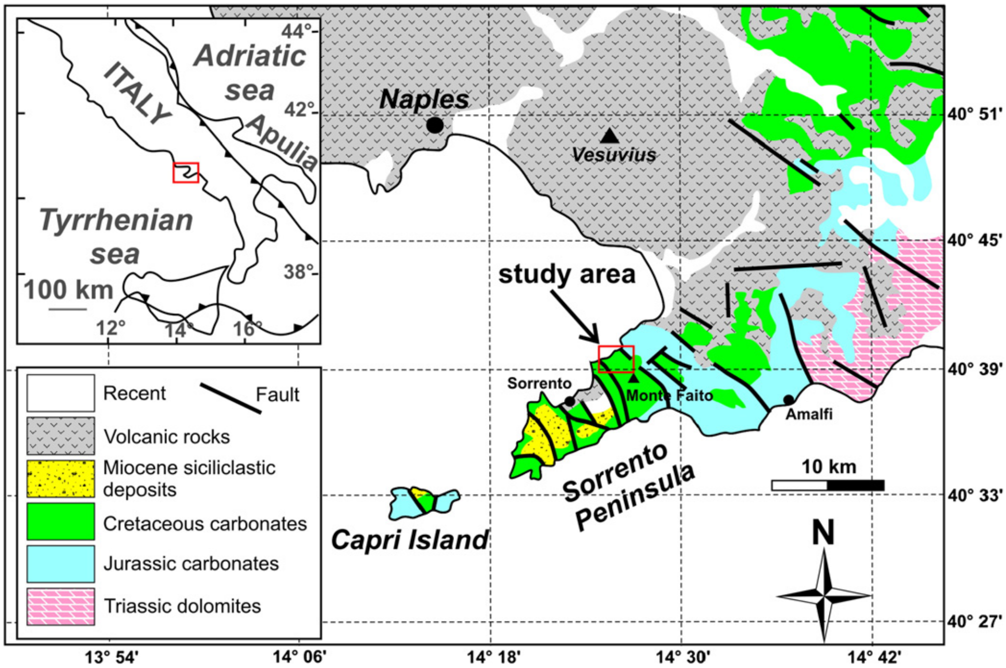

Fig. 2. Geological sketch map of the southern Apennines in the Naples area, showing location of field study area. After Guerriero et al., 2010.

water carbonate succession of the so-called Apennine Platform (e.g. Cello and Mazzoli, 1999; Mazzoli et al., 2001, 2008; Corrado et al., 2005, and references therein). The analyzed beds are part of a 50 m thick pack of strata that has been studied in detail also in terms of sedimentology, petrography, geochemistry and petrophysical characteristics (Galluccio et al., 2008). Statistical sampling has been performed from 2 to 4 m long scan lines at outcrop, as well as from *ca.* 15 cm long micro-scan lines by means of a digital camera (Proscope, used at outcrop or on oriented polished slabs) at a magnification level of 50×, and micro-scan lines on oriented thin sections by optical microscopy at a magnification of 200×.

2. Fracture data collection

The presented statistical analysis is mainly aimed at the study of non-stratabound fracture sets. Statistical sampling has been carried out by recording fractures intersecting suitable oriented ideal lines (scan lines) that are drawn along outcrop surfaces, polished slabs, and thin sections. Eight mechanical strata, characterized by homogeneous lithology and texture, have been analyzed along bedding-parallel scan lines. Rock types include fine-grained limestones (mudstones; beds 1, 56, and 107), as well as medium- (beds 66 and 71) and coarse-grained (beds 102, 118, and 120) dolomites. Fracture type (vein or joint), distance from the origin, attitude, length, aperture, morphology, crosscutting relationship, composition and texture of fracture fill have been recorded for each fracture, together with mechanical layer thickness. Fracture aperture (i.e. opening displacement) has been evaluated by means of a logarithmically graduated comparator (Ortega et al., 2006). Three micro-scan lines have been performed by means of the Proscope digital camera: one at outcrop (on a mudstone layer, bed 56) and two on oriented polished slabs (medium-grained dolomite, bed 66; and coarse-grained dolomite, bed 118). Finally, micro-scan lines have been carried out on oriented thin sections of mudstones (beds 56 and 107) and medium-grained dolomite (bed 66). As microstructural analysis was mainly aimed at the study of fracture aperture and spacing, the parameters of fracture type (vein or joint), distance from the origin, and fracture aperture have been recorded for each fracture set intersecting the micro-scan lines.

The analysis of fracture attitudes reveals the occurrence of two main fracture sets (Guerriero et al., 2010), at a high angle to each other and orthogonal to bedding, here conventionally called set 1 (dip angle: *c.* 80°, dip direction: 10° to 45°) and set 2 (dip angle: *c.* 70°, dip direction: 300° to 340°). Statistical analysis has been performed separately for each of the two sets, including joints and veins. Table 1 summarizes the collected data set.

3. Statistical analysis

In the following sections, new, effective methods for multi-scale statistical analysis of fracture spatial distribution and aperture probability distribution are outlined.

3.1. Analysis of spatial fracture distribution

In order to establish the type of statistical distribution for fracture spacing, mean value and standard deviation have been plotted vs. sample size (i.e. number of fractures; Fig. 3). In all cases, the standard deviation converges towards the mean spacing value. Therefore, the coefficient of variation (the ratio between standard deviation and mean value; see Gillespie et al., 1999) systematically tends to unity. This feature is typically associated with a uniform random spatial distribution of fractures (Gillespie, 2003; Gillespie et al., 1999, 2001).

Based on probabilistic theory (e.g. Balakrishnan and Basu, 1996; Dekking et al., 2005), a uniform random spatial distribution of structural features (belonging to the same set) is expected to produce a fracture spacing distribution following an exponential law. Moreover, fracture density follows a Poisson's distribution. The exponential random variable (RV) shows the following main characteristics (Balakrishnan and Basu, 1996): (a) its mean value distribution conforms to a Chi squared RV, and (b) its mean value and standard deviation attain the same value. Because of this latter condition, the convergence to unity of the coefficient of variation of spacing values may lead one to hypothesize an exponential distribution for fracture spacing and, consequently, a random spatial distribution of fractures. Nevertheless, such a convergence is a necessary but not sufficient condition for inferring a uniform fracture spatial distribution, since

Table 1
Summary of fracture data sets collected at different scales.

Bed N. and rock type	Outcrop scan line	Proscope (50×)	Thin section (200×)
	N. of fractures	N. of fractures	N. of fractures
Bed 1 — set 2 mudstone	75	–	–
Bed 56 — set 1 Mudstone	52	33	34
Bed 56 — set 2 Mudstone	10	–	46
Bed 66 — set 1 Medium dolomite	18	14	<10
Bed 66 — set 2 Medium dolomite	23	20	–
Bed 71 — set 1 medium dolomite	15	–	–
Bed 71 — set 2 Medium dolomite	28	–	–
Bed 102 — set 1 Coarse dolomite	29	–	–
Bed 102 — set 2 Coarse dolomite	40	–	–
Bed 107 — set 1 Mudstone	16	–	20
Bed 107 — set 2 Mudstone	18	–	14
Bed 118 — set 2 Coarse dolomite	54	<10	–
Bed 120 — set 2 Coarse dolomite	54	–	–

several distributions can exhibit a unitary coefficient of variation. On the other hand, natural fracture systems may indeed be characterized by a uniform random spatial distribution, but in some cases the statistical sample could not be large enough to appreciate the convergence of the coefficient of variation toward unity. In order to ascertain that joints are randomly distributed, in the present study the analysis is carried out following a series of steps. First, the bootstrap method is applied to spacing data in order to determine whether mean spacing distribution conforms to a Chi squared RV. Second, probability plots are employed to verify that spacing values display an exponential distribution. Finally, the stochastic independence of consecutive spacing values (required for a random spatial distribution of fractures) is verified by means of autocorrelation analysis.

Once the types of statistical distributions for fracture spacing and density are defined, the probability distributions are defined by providing mean value and standard deviation. Nevertheless a more complete characterization of such distributions requires the quantification of uncertainties associated with mean sampling estimate. This can be obtained by calculating the 95% confidence interval (defined as the real interval containing the population mean with probability of 95%) of mean fracture density and mean spacing. This interval can be rapidly calculated by means of the following equations (Guerriero et al., 2010):

$$F_{low} = 1 - \frac{1.96}{\sqrt{N-1}} \cdot \frac{N}{L} \tag{1}$$

$$F_{upp} = 1 + \frac{1.96}{\sqrt{N-1}} \cdot \frac{N}{L} \tag{2}$$

where F_{low} and F_{upp} indicate the limits of the confidence interval for fracture density (as well as for spacing), N is the total number of measured fractures, and L is the scan line length. Eqs. (1) and (2) introduce an approximation in the calculus, according to the central limit theorem. Comparing these results with those calculated by means of an exact method involving the Poisson probability distribution for the fracture density, it follows that such equations furnish sufficiently accurate results for N≥20. Furthermore, for a first approximation estimate, a value of N≥10 is sufficient.

Finally, we wish to emphasize the usefulness of the proposed analysis criterion in those cases in which, although fractures do not show a random distribution, spacing distribution is well approximated by an exponential distribution (i.e., the first two steps above give positive results, but the stochastic independence of consecutive spacing values is not verified). In fact, in these instances the application of Eqs. (1) and (2) is still appropriate.

3.1.1. Bootstrap method

The bootstrap method (e.g. Dekking et al., 2005; Wasserman, 2006) has been applied to spacing data. For each scan line data set, the mean of ten randomly chosen spacing values has been calculated. This process has been repeated 10,000 times, obtaining 10,000 mean values. Subsequently, the cumulative distributions of these sampling mean values have been calculated. Let us now consider the following function: $y = \frac{2n}{\mu}x$, where x is the mean of n determinations of an exponential RV, and μ is the mean of the whole population of such an RV (i.e. the 'true' mean). Since y shows a Chi squared distribution (e.g. Balakrishnan and Basu, 1996), the inverse function of this distribution has been applied to the observed cumulative distributions. It has then been verified whether the calculated values fall along a line of equation: $y = \frac{2N_f}{S}x$, where N_f is the number of spacing values and S is the mean spacing (note that S is calculated from the whole data set, whilst x is calculated on ten values). This has been carried out using a diagram where the mean (x) is plotted along the abscissa axis and the inverse Chi squared function along the ordinate axis (Fig. 4). Furthermore, the inverse Gauss function has been applied to the same data set. The diagrams in Fig. 4 show that data points are well aligned

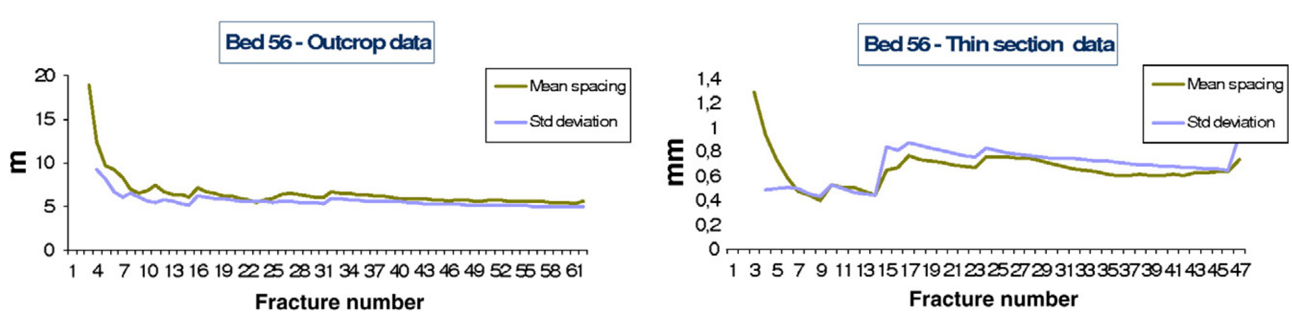

Fig. 3. Diagrams of mean and standard deviation of fracture spacing in limestone (mudstone, bed 56). Data from outcrop- and micro- (microscope at a magnification of 200×) scan lines are plotted separately. Note how mean spacing and standard deviation converge towards the same value.

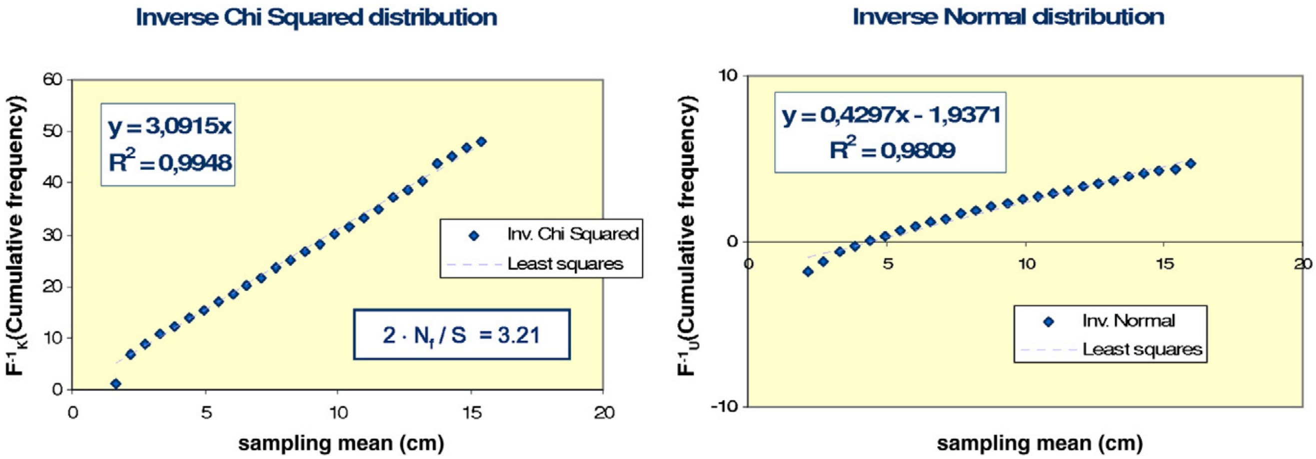

Fig. 4. Bootstrap probability plots for Chi squared and Normal distributions of fracture spacing (see text for details). Note that, for the inverse Chi squared function, the slope of the least squares line is consistent with the theoretical slope given by: $\frac{2N_f}{S}$.

around the mean squares line, however they show a systematically upward convexity for the inverse Gauss distribution. On the other hand, they show non-systematic deviations for a Chi squared distribution. These features indicate that mean spacing is characterized by a Chi squared distribution; however, the Gauss distribution furnishes a good approximation already for mean values calculated from 10 fractures (thus justifying the approximation made in Guerriero et al., 2010, in order to obtain Eqs. (1) and (2)).

3.1.2. Probability plots

For each data set, spacing values have been plotted against the inverse distribution function of the observed spacing cumulative distribution according to the exponential probability model (Fig. 5). These diagrams may be used to verify whether an analyzed RV is characterized by a certain probability distribution and, if this is the case, to obtain the equation of the theoretical distribution best fitting the data. If the RV is exponential with a mean value μ, then the cumulative distribution approaches to an exponential curve of equation $y = 1 - e^{-x/\mu}$ (i.e. the function $1 - y$ forms a line in a diagram with logarithmic ordinates). For most of the data set analyzed in this study, an exponential distribution provides a good fit of observed data points, at both outcrop- and micro-scale.

3.1.3. Autocorrelation analysis of consecutive spacing values

A uniform random spatial distribution of fractures yields an exponential distribution of spacing values. Nevertheless, the opposite is not generally true. This concept can be clarified by means of a simple example: consider a randomly distributed fracture set yielding a sample of *N* spacing values and imagine sorting them by increasing values from left to right in a diagram. Although spacing distribution is again exponential, the fracture spatial distribution is no longer uniform, as there is a higher fracture density on the left side of the diagram (i.e. it is more probable to encounter a fracture on the left than on the right side of the diagram). In order to ascertain a uniform random spatial distribution of fractures, once an exponential distribution of spacing values has been verified, it must also be proved that consecutive spacing values are stochastically independent. A useful criterion used in geostatistics for evaluating the dependence between consecutive values attained by an RV consists in the analysis of the variance, covariance or correlation of elements put at increasing distance (Diggle and Ribeiro, 2007). In this paper, the correlation between contiguous values of spacing has been analyzed (note that covariance analysis provides similar results; Diggle and Ribeiro, 2007). The first correlation value is calculated between the nth and n + 1 spacing values, for each n. Successive values of correlation are calculated between nth and n + 2, … n + k, … values etc. Then the calculated correlation values are plotted versus the variable *k* (correlogram). In order to clarify the usefulness of this method, correlograms of two irregular time series are represented in Fig. 6. Such series, produced by Monte Carlo simulation, are of the type: (a) $y(n+1) = y(n) + g(\sigma)$; and (b) $y(n) = g(\sigma;)$ where $g(\sigma)$ is a Gaussian RV, independent from position, with mean equal to zero and constant variance σ, and $y(n)$ are the consecutive values attained by the RV (in this case spacing). In a type a series, each value is a linear function of the preceding one (therefore they are linearly correlated). The related correlogram shows a correlation value closely approaching unity for k = 1 (and for small k values), gradually decreasing as k increases (Fig. 6a). In a type b series, each value is stochastically independent of the preceding one (i.e. they are uncorrelated). In this case the correlation attains very low values, with respect to unity, for any value of k. The related correlogram shows a very irregular trend (Fig. 6b).

Examples of correlograms of spacing data from analyzed outcrop and micro-scan lines, for k values in the range [1; 15], are shown in Fig. 7. In all cases the correlation attains low values (even for small values of k) and a rather irregular trend, pointing out a lack of statistical correlation between each spacing value and the contiguous ones. Should be noted that a robust correlation analysis requires use of adequately large samples. Establishing the minimum sample size, required for a meaningful analysis, goes beyond the scope of this paper and could be the object of future research. In this study we utilized statistical samples including at least twenty fractures.

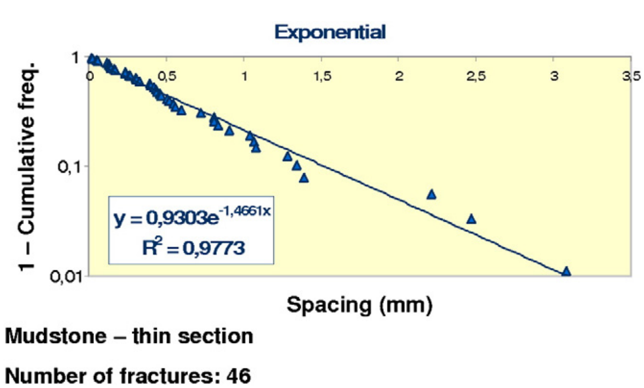

Fig. 5. Probability plot for the recorded cumulative distribution of spacing values.

3.2. Analysis of fracture aperture probability distribution

Multi-scale statistical analysis confirms that, in fine-grained rocks, the aperture cumulative frequency follows a power law over several

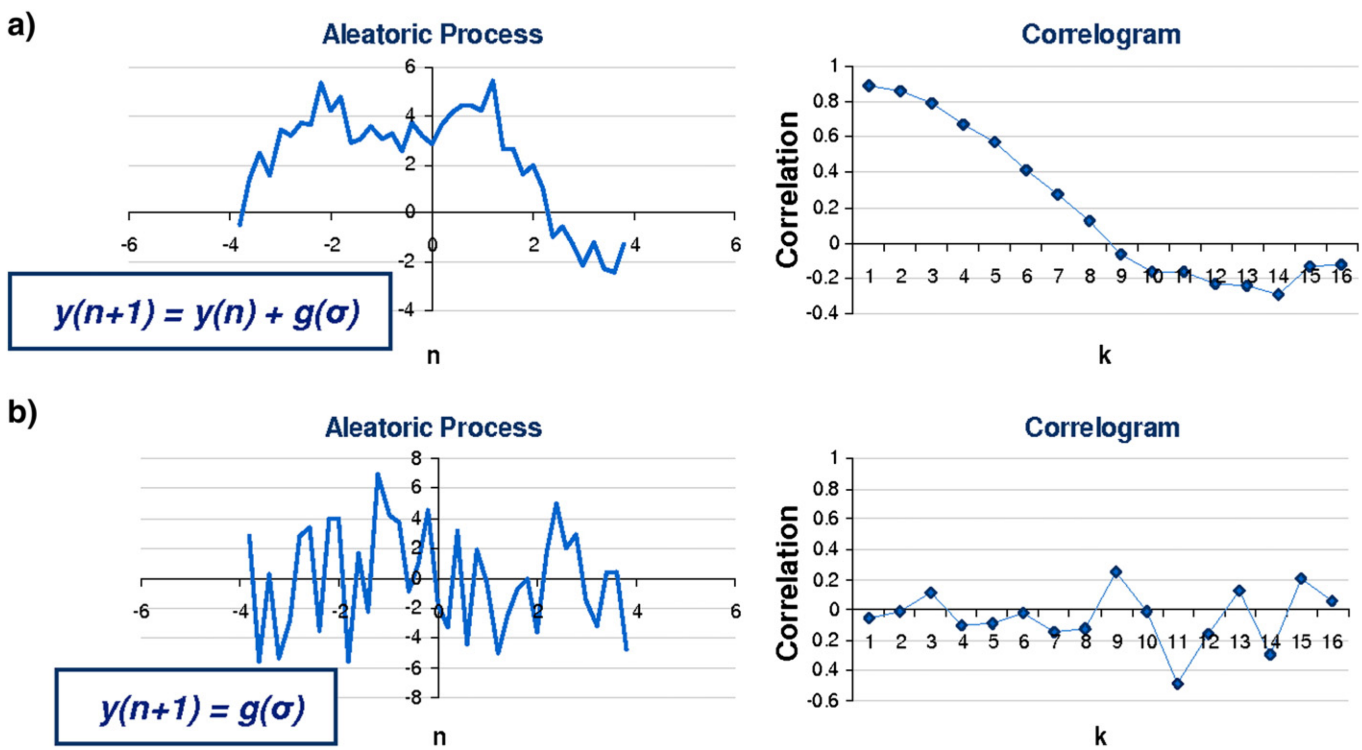

Fig. 6. Aleatoric processes and related correlograms. (a) Process (or time series) in which each value is correlated with the neighboring ones. (b) Process characterized by uncorrelated consecutive values.

orders of magnitudes (Fig. 8), in good agreement with (e.g.) Ortega et al. (2006) (the cumulative frequency of an aperture value *b* is defined here as the number of joints per meter having aperture greater than b). However, fracture behavior at the micro-scale deserves a deeper analysis. In micritic limestones (grain size below 5 µm) the power law appears to be valid down to the thin section scale, i.e. for apertures of few microns. The estimated maximum densities are larger than 1700 fractures per meter (Fig. 8). On the other hand, for medium-grained dolomites (grain size in the range of 20–80 µm) fracture density reaches about 200 fractures per meter (Fig. 8). For these rocks, only micro-scan lines at a magnification of 50× (by means of the Proscope digital microcamera) could be performed, due to the lack of a significant number of fractures in thin-section (200×). The problem of micro-scale sampling is even more evident for coarse-grained dolomites (grain size > 120 µm). For these rocks, scan lines performed at outcrop furnish density estimates of a few tens of fractures per meter, whereas micro-scan lines, performed by Proscope or, even more so, in thin section, do not provide a statistically meaningful number of fractures. These results emphasize the fundamental role of grain size (as opposed to truncation artifacts; e.g. Ortega et al., 2006) in controlling the lower validity limit of the power-law distribution, as well as maximum fracture density. The statistical analysis described below is focused on the range of aperture cumulative distribution for which a power law can be considered valid.

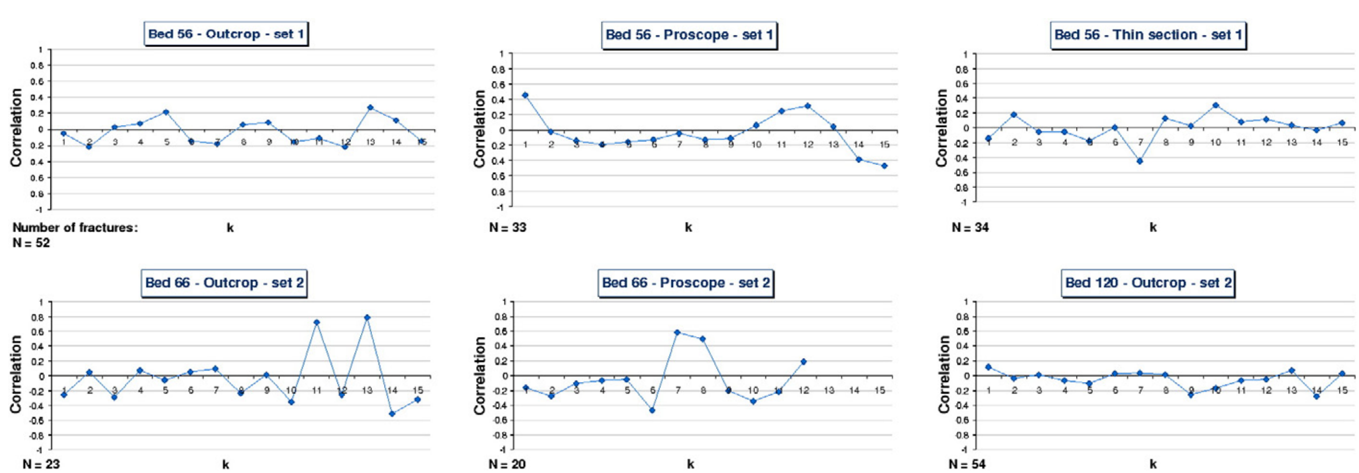

Fig. 7. Calculated correlograms of scan line data from the three different carbonate types analyzed in this study (see text for details). The diagrams show data collected at various scales.

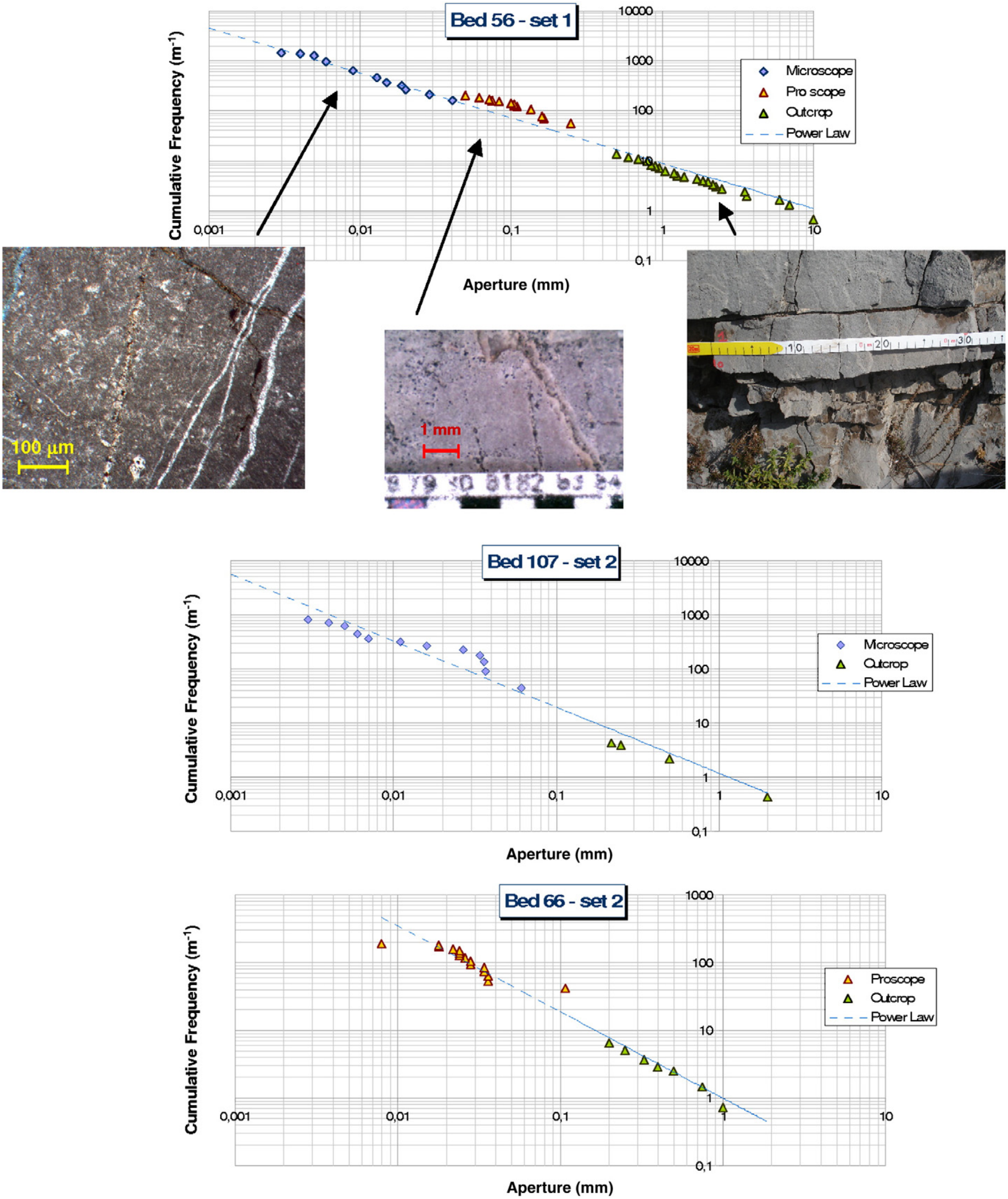

Fig. 8. Cumulative frequencies of fracture apertures, obtained by the integration of outcrop- and micro-scale scan line data on mudstones (beds 56 and 107) and medium-grained dolomite (bed 66). The curves follow a power law over several orders of magnitude.

3.2.1. Uncertainties and errors in scan line statistics

Bonnet et al. (2001) point out that use of cumulative distribution has some drawbacks, suggesting to utilize frequency density distribution (i.e. the derivate of cumulative distribution). Nevertheless use of this latter distribution requires large data set which, in many cases, may be not available. Another question related to cumulative distribution consists in establishing how many fractures are needed in order to achieve a meaningful estimate of the power law exponent (Bonnet et al., 2001). In next section we show a method which overcomes these issues. Before describing this method we illustrate in detail the main problems associated with use of cumulative frequency.

The power law approximating the cumulative frequency of fracture apertures is usually determined by calculating the least squares line interpolating the size-frequency data (an alternative criterion is the maximum likelihood method which is addressed in Section 3.2.4). The observed cumulative frequency of fracture apertures consists of fracture density estimates calculated for each aperture value. The uncertainty related to each mean fracture density value can be described by means of the 95% confidence interval, which can be obtained by Eqs. (1) and (2). As each confidence interval is characterized by amplitude being a function of the same estimated value (i.e. fracture density N/L) it follows that, as shown in Fig. 9, the standard deviation varies along the abscissa (heteroskedasticity, Dekking et al., 2005). Therefore, the application of the least squares method is inappropriate without the application of appropriate weight function for residuals. The main problem of this method consists of a large aleatoric error in estimating the slope of the least squares line (i.e. the exponent of the power law). This is due to the larger confidence intervals in the right-hand side of the data distribution with respect of those on the left-hand side. A significant diminution of the uncertainty on the power-law exponent can be obtained by means of multi-scale analysis, i.e. by the integration of data from micro-scan lines with the traditional analysis performed by means of scan lines at outcrop (Fig. 8). Multi-scale analysis has been applied successfully by many authors for scan lines and scan areas (Bonnet et al., 2001, and references therein; Bour et al., 2002; Davy et al., 2006; Odling, 1997; Ortega et al., 2006). However, the problem of the applicability of the least squares method holds also in this case.

3.2.2. Uncertainties reduction criterion in multi-scale statistical analysis

The aforementioned problem can be overcome by an analysis based on the definition of a single line passing for two points only, i.e. one for each — outcrop and micro-scale — data set (Fig. 10). These points are chosen in such a way as to obtain the maximum number of sampled fractures for each data set, these being affected by the minimum truncation error (e.g. Bonnet et al., 2001; Ortega et al., 2006). Based on our experience and practice, a lower aperture threshold of 0.2 mm (Ortega et al., 2006) is a quite small size for naked-eye measurements, and could originate some truncation artifacts. Therefore, we suggest a fracture aperture threshold of 0.5 mm for outcrop-scale fracture analysis. For micro-camera images, taking into account digital microcamera (Proscope) image resolution (640×480 pixels), we propose a lower limit of 0.05 mm, whereas for thin section images under the optical microscope we propose a threshold of 0.005 mm. In analyzing data, slightly lower threshold values can be adopted (e.g. if a data set from outcrop scan line is quite poor, we could fix a threshold value to 0.2 mm) nevertheless, in so doing, one should be aware that although the uncertainty is reduced, the risk of truncation artifact increases. Such an approach permits a significant reduction of the aleatoric error in estimating the power-law exponent, as shown in the following section.

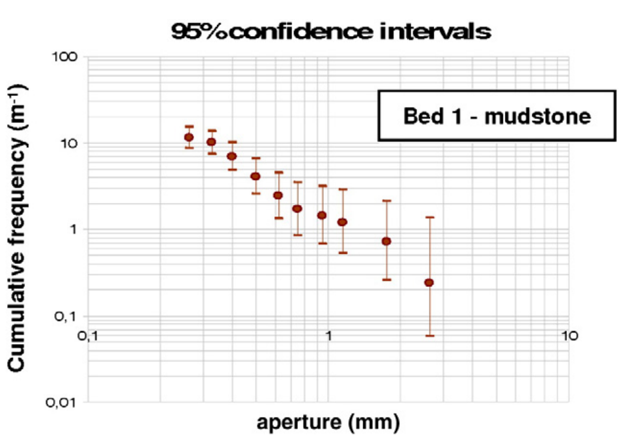

Fig. 9. Diagram showing confidence intervals for the recorded frequency values of aperture data from outcrop-scale scan line (mudstone, bed 1). Note how confidence intervals can vary largely along the abscissa axis (heteroskedasticity), producing significant uncertainties in estimating the slope of the least squares line.

3.2.3. Monte Carlo simulation

In order to test the method described in the previous section, Monte Carlo simulations (e.g. Dekking et al., 2005; Good, 2006) have been performed. The experiment simulates a set of sampling estimates of the power-law exponent and coefficient, according to the three methods shown in Fig. 11. The applied theoretical distribution is a power law characterized by reliable exponent and coefficient values, equal to 1.1 and 7.5, respectively. Furthermore, random spatial fracture distributions over a 4 m long scan line and a 15 cm long micro-scan line have been assumed. Each simulation produces an empirical aperture cumulative distribution according to the following criterion. In a scan line of length L, the number of detected fractures belonging to a certain aperture class is a Poisson RV, whose mean is given by the power law. For example, the mean fracture number per meter whose aperture is in the range of 0.5 to 0.75 mm is 1.97 (provided by the power law with exponent and coefficient equal to 1.1 and 7.5). Thus, in a 4 m long scan line the expected fracture number is 7.87 (given by: $1.97 \cdot 4$). The detected fracture number is a Poisson RV with a mean value equal to 7.87. Applying the Monte Carlo method (e.g. Dekking et al., 2005; Good, 2006), a determination of this latter RV (i.e. the detected fracture number) is simulated, and such a process is repeated for any considered aperture class. The final result is an artificial cumulative distribution of aperture values. The simulation has been repeated 10,000 times. In the simulation of a traditional scan line data set, only fractures with apertures larger than 0.5 mm have been considered in the calculation of the best-fit power law. In the case of multi-scale analysis, aperture values in the range 0.05–0.5 mm, for micro-scan lines, and larger than 0.5, for outcrop-scale scan lines, have been considered. Finally, the least squares line and, for multi-scale analysis, also the line passing for two points (with abscissa values of 0.05 and 0.95) have been calculated. In this way we produced 10,000 simulated sampling estimates of the power-law exponent and coefficient. The results of the Monte Carlo simulations are shown in Fig. 11a, as bivariate probability distributions of coefficient and exponent estimates. These diagrams show that the two-point fitting method provides a smaller dispersion of estimates with respect to the least squares method. This is confirmed by the analysis of the univariate probability distribution of the power-law exponent (Fig. 11b). In order to emphasize the significant advantage provided by the use of this criterion, an example is addressed here. For a data set sampled by a 4 m long scan line and a 15 cm micro-scan line (considering the theoretical power law described above), the power-law exponent showed a standard deviation equal to 0.77. In order to obtain the same standard deviation by use of the least squares criterion, we would need to carry out a 12 m and 30 cm sized scan line and micro-scan line, respectively (Table 2).

3.2.4. Comparison with the maximum likelihood method

A more effective criterion than the least square method, aimed at estimating the power-law exponent, is the maximum likelihood method. The basic theory of this method is thoroughly illustrated in Newman (2005), where practical equations are also provided to obtain the power-law exponent estimate and its confidence interval. In statistical inference, the maximum likelihood method generally provides very robust and highly converging estimators. Although this

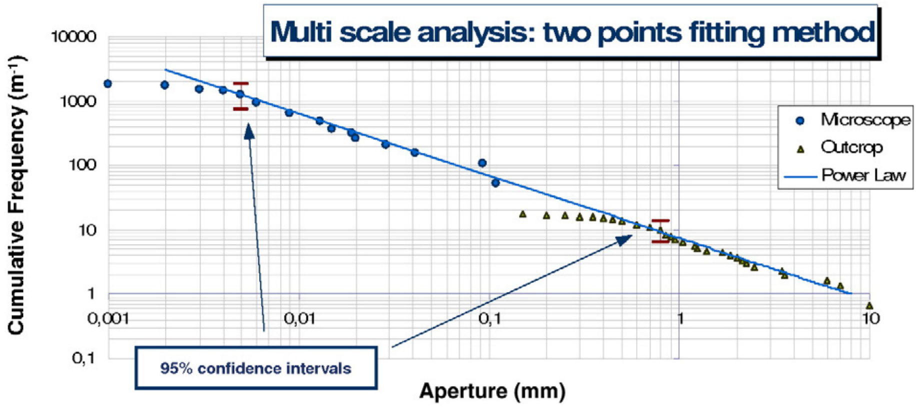

Fig. 10. Diagram showing application of the two-point fitting method proposed in this study. A power-law distribution can be individuated promptly by using a line for two opportunely chosen points, rather than a least-squares line. For each data set, the points on the left side of the curve tend to align along a sub-horizontal trend as a result of truncation artifacts. On the other hand, the points to the right are characterized by ordinate frequency estimates that are based on a less numerous sample — being therefore less accurate — than those to the left. Therefore, for each data set, the points are chosen as much as possible on the left side of the curve, but before this attains a sub-horizontal trend.

method provides a more efficient estimator than the least squares method also in analyzing power-law distributions, the formulation for estimating the exponent and confidence interval provided in Newman (2005) cannot be applied to a multi-scale data set like that utilized in this paper. Considering the example described in the previous section, in order to achieve a standard deviation of 0.77 for the exponent estimate by using the maximum likelihood method with outcrop-scale scan line data, a much longer scan line would be needed. Indeed the maximum likelihood standard deviation of the exponent estimate m is given by $\frac{m}{\sqrt{N}}$, where N is the number of sampled fractures (Newman, 2005). By considering $\frac{m}{\sqrt{N}} = 0.77$, we obtain $N \approx 200$ (being $m = 1.1$). Supposing that only fracture aperture values greater than 0.5 mm are checked, one would have to perform a

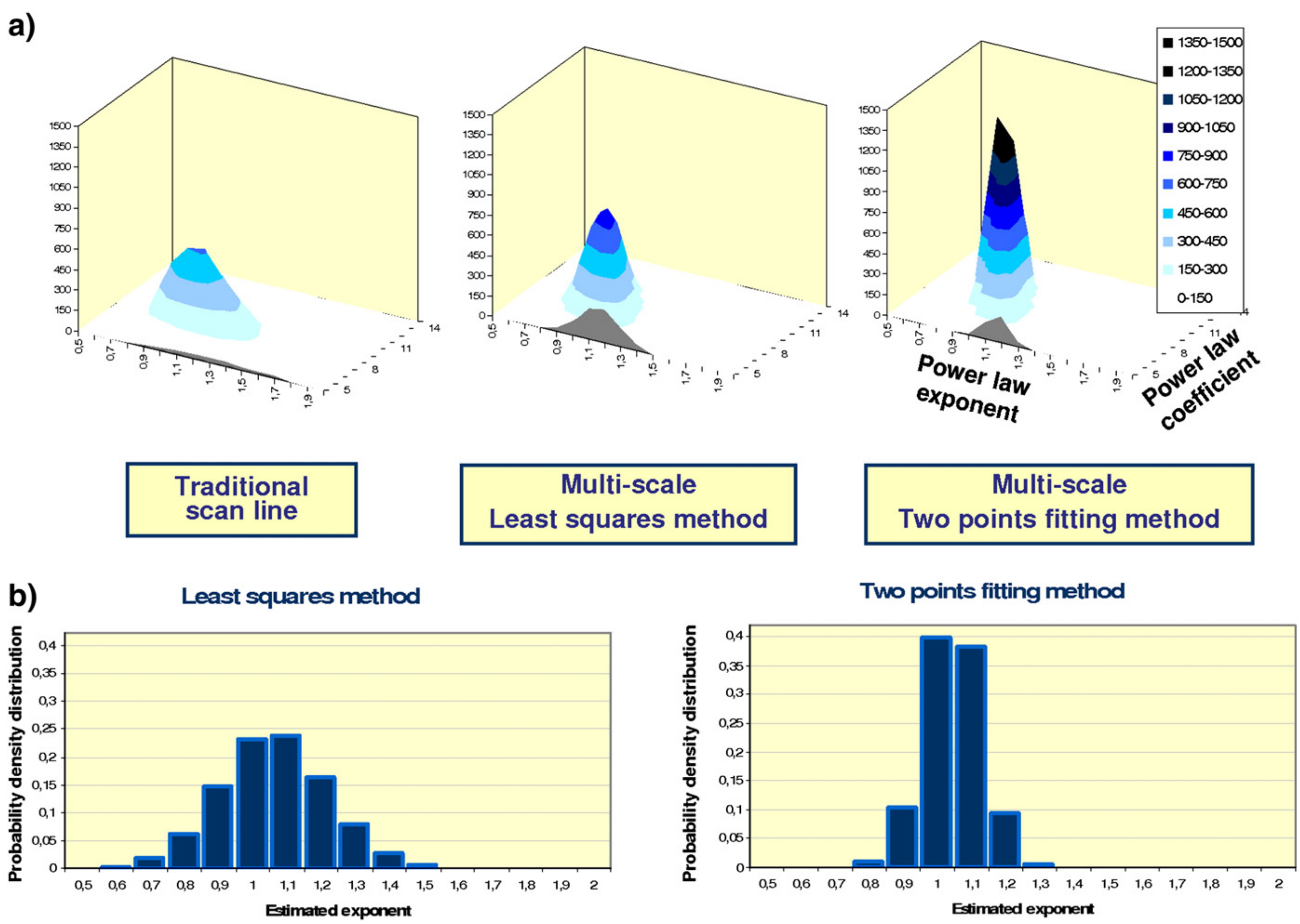

Fig. 11. Monte Carlo simulations modeling the estimation of exponent and coefficient of the power law for fracture apertures (10,000 determinations). (a) Bivariate distributions of power-law coefficient and exponent. (b) Comparison between distributions of the sole exponent for least squares and two-point fitting methods. The simulation points out how the two-point fitting method reduces the dispersion — and therefore the uncertainty — of estimated coefficient and exponent, with respect to the least squares criterion.

Table 2
Comparison among various criteria used to estimate the exponent of the power-law distribution. Note that the two-point fitting method requires the minimum scan line and microscan line length to achieve the same value of standard deviation for the exponent.

Method	Standard deviation of the exponent estimate	Scan line length (m)	Micro-scan line length (cm)
Traditional scan line	0.078	20	–
Multi-scale scan line least squares	0.076	12	30
Maximum likelihood	0.078	12.5	–
Multi-scale scan line two-point method	0.078	4	15

12.5 m long scan line in order to sample such a number of fractures ($N = 200$). The results obtained by the described methods are compared in Table 2.

3.2.5. Confidence interval for the power-law exponent

The proposed two-point fitting method, besides reducing the error in estimating the power-law exponent, furnishes a rapid way to calculate the confidence interval of the power-law exponent. Indeed, the real values of fracture density, for the two points, fall in the respective 95% confidence intervals (Fig. 12). Furthermore, also the real power law (represented by a line in a bi-logarithmic diagram) falls within one of these confidence intervals, with the same probability. The joint probability that the line falls within the two intervals simultaneously is equal to the product of the probabilities that the line passes through each interval, being the two data set (quasi) stochastically independent (note that outcrop- and micro-scale scan lines have very few common points). Therefore, the final probability is equal to 0.95^2, i.e. *ca.* 0.9.

By indicating with F_{low1} and F_{upp1} the confidence interval limits of the outcrop-scale scan line, F_{low2} and F_{upp2} those relative to the micro-scan line, s1 and s2 the abscissas of the two points and applying Eqs. (1) and (2), the following equations for the 90% confidence interval of the exponent m have been obtained:

$$m_{upp} = \frac{ln\left(\frac{F_{low1}}{F_{upp2}}\right)}{ln\left(\frac{s_1}{s_2}\right)} \quad (3)$$

$$m_{low} = \frac{ln\left(\frac{F_{upp1}}{F_{low2}}\right)}{ln\left(\frac{s_1}{s_2}\right)}. \quad (4)$$

The described method has been applied to the real data set illustrated in Fig. 8. For these multi-scale scan line data from fractured carbonates, the power-law exponent and the related confidence interval attain the following values:

- bed 56 (mudstone) set 1: $m = 0.95$; $m_{upp} = 1.1$; $m_{low} = 0.8$
- bed 107 (mudstone) set 2: $m = 1.16$; $m_{upp} = 1.49$; $m_{low} = 0.9$
- bed 66 (medium dolomite) set 2: $m = 1.44$; $m_{upp} = 1.93$; $m_{low} = 0.92$

The illustrated criterion has general validity and can be usefully applied in all that cases in which a studied RV is distributed in accordance to a power law (e.g. Newman, 2005), requiring the only hypothesis that each cumulative frequency value follows a Poisson distribution in the sampling domain.

4. Concluding remarks

A methodology of statistical analysis of multi-scale scan line data from fractured reservoir analogues is illustrated, which yields an enhanced definition of spatial tensile fracture distribution as well as of fracture aperture probability distribution. This methodology has been applied to fractured carbonates of the Sorrento Peninsula, which represent an analogue of buried reservoirs of southern Italy's major oil fields.

Concerning fracture spacing, the integrated application of the bootstrap method, probability plots and autocorrelation analysis confirms a uniform random spatial distribution of fractures. Regarding fracture aperture, multi-scale statistical analysis confirms that its cumulative frequency conforms to a power law over several orders of magnitudes. However, grain size exerts a fundamental control on both maximum fracture density and lower validity limit of the power-law distribution.

A novel criterion has been introduced for aperture multi-scale analysis, based on the definition of a single line passing for two points only (i.e. one for outcrop- and one for micro-scale data sets) rather than the least squares method. The use of this criterion provides notable advantages, as:

(1) it significantly increases the precision of the power-law exponent estimate (as effectively proved by Monte Carlo simulations);

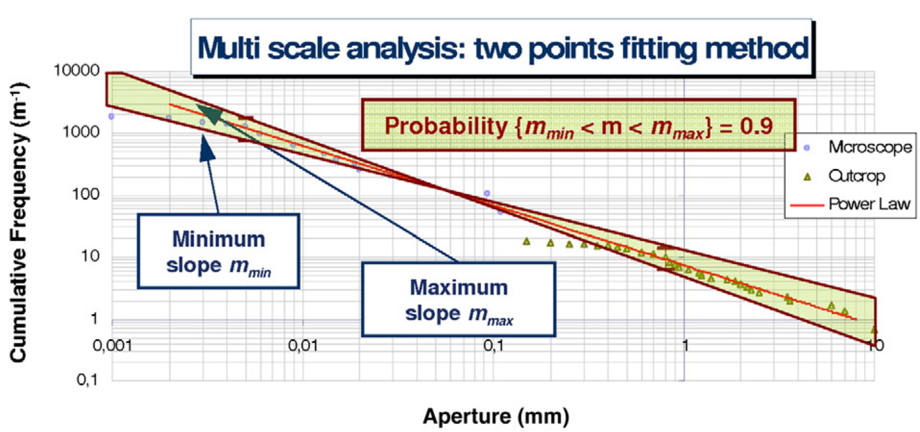

Fig. 12. Maximum and minimum slope value for the power-law distribution of fracture apertures. The line is contained within each of the two confidence intervals with a probability of 95%. The product of the two probabilities furnishes (with a good approximation) the probability that the line is contained simultaneously within the two intervals, this being equal to $0.95^2 \approx 0.9$.

(2) it allows deriving new expressions, in closed form, for the limits of the confidence interval of the exponent estimate;
(3) it allows to effectively overcome the problems associated with the use of cumulative distributions (see Section 3.2.1); and
(4) it has a general validity and can be applied to several scientific contexts in which the analyzed RV shows a power-law probability distribution.

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